

Comparative Evaluation Report

Water-use and Electricity ODD+D action-situation extraction: ODD+game context vs ODD-only

Field	Value
Report date	2026-05-11
Workspace	/Users/alex-lirio-lucian/develop/WaterUse-alex/LLM_Prompting
Cases	ODD_water and ODD_electricity
Evaluation outputs	Current comparison reports and AS-level audit CSV files in each case folder

Executive summary

The ODD+game condition is not uniformly better across all models and cases. It improves recall for DeepSeek and Qwen in several comparisons, but it can also increase over-generation, duplicates, or missed specific ASs. The most reliable interpretation comes from TP, FN, FP, Precision, and Recall together, plus AS-level audit records.

1. Test Environment

Item	Configuration
Provider/API	Together API via together Python client
Model count	Three LLMs evaluated in each case and prompt condition
Models	DeepSeek-R1; Llama-3.3-70B-Instruct-Turbo; Qwen2.5-7B-Instruct-Turbo
Prompt conditions	ODD+game: ODD+D text plus game scenario; ODD-only: ODD+D text only
Runs	30 runs per model per condition; 180 total generations per case
Generation settings	temperature=0.6; max_tokens=16000; timeout=600 seconds for all three models
Water-use scorer	ODD_water/evaluate_tp_fp_fn_compare.py
Electricity scorer	ODD_electricity/evaluate_tp_fp_fn.py

2. Evaluation Method

Each generated action situation was reviewed against a fixed gold set for the relevant case. The scoring scripts evaluate only TP, FN, FP, Precision, and Recall. F1 is intentionally not included. Titles are not ignored: the generated title is judged together with the body and payoff matrix or sequential representation text.

Metric	Definition
TP	Generated AS is in the correct AS set. The first match to a correct AS counts as TP.
FN	Correct AS was missed by the model in that run.

Metric	Definition
FP	Generated AS is not in the correct set, or is a duplicate after that correct AS was already counted.
Precision	$TP / (TP + FP)$
Recall	$TP / (TP + FN)$

2.1 Correct AS Sets

Water-use correct set

Water AS	Correct action situation
AS1	Upstream and downstream withdrawal decisions
AS2	Fish extraction common pool resource game

Electricity correct set

Electricity AS	Correct action situation
AS1	Capacitor-adoption assurance game between neighbouring farmers
AS2	Sequential social-learning process in capacitor adoption
AS3	Asymmetric transformer-capacity authorization dilemma between farmers
AS4	Mutual-exchange coordination game between farmer and sub-station staff
AS5	Authorization-and-investment asymmetric coordination between farmer and staff
AS6	Groundwater-extraction prisoner's dilemma between farmers

3. Water-use Results

The water-use case has two confirmed correct ASs, so each model has a maximum of 60 TP over 30 runs. The comparison shows that ODD+game improved recall for DeepSeek-R1 and Qwen2.5-7B, but not for Llama-3.3-70B.

Table 1. Water-use aggregate results

Model	Setting	TP	FN	FP	Precision	Recall
DeepSeek-R1	ODD+game	49	11	10	0.8305	0.8167
DeepSeek-R1	ODD-only	46	14	9	0.8364	0.7667
Llama-3.3-70B	ODD+game	54	6	62	0.4655	0.9000
Llama-3.3-70B	ODD-only	55	5	53	0.5093	0.9167
Qwen2.5-7B	ODD+game	39	21	52	0.4286	0.6500
Qwen2.5-7B	ODD-only	36	24	58	0.3830	0.6000

Table 2. Water-use deltas: ODD+game minus ODD-only

Model	Delta TP	Delta FN	Delta FP	Delta Precision	Delta Recall
DeepSeek-R1	+3	-3	+1	-0.0059	+0.0500
Llama-3.3-70B	-1	+1	+9	-0.0437	-0.0167
Qwen2.5-7B	+3	-3	-6	+0.0456	+0.0500

4. Electricity Results

The electricity case has six confirmed correct ASs, so each model has a maximum of 180 TP over 30 runs. ODD+game gave DeepSeek-R1 perfect recall and fewer FP than ODD-only. Llama-3.3-70B gained a small precision advantage but lost recall. Qwen2.5-7B achieved perfect recall but produced many extra ASs, lowering precision.

Table 3. Electricity aggregate results

Model	Setting	TP	FN	FP	Precision	Recall
DeepSeek-R1	ODD+game	180	0	4	0.9783	1.0000
DeepSeek-R1	ODD-only	164	16	12	0.9318	0.9111
Llama-3.3-70B	ODD+game	166	14	3	0.9822	0.9222
Llama-3.3-70B	ODD-only	176	4	4	0.9778	0.9778
Qwen2.5-7B	ODD+game	180	0	103	0.6360	1.0000
Qwen2.5-7B	ODD-only	177	3	52	0.7729	0.9833

Table 4. Electricity deltas: ODD+game minus ODD-only

Model	Delta TP	Delta FN	Delta FP	Delta Precision	Delta Recall
DeepSeek-R1	+16	-16	-8	+0.0464	+0.0889
Llama-3.3-70B	-10	+10	-1	+0.0045	-0.0556
Qwen2.5-7B	+3	-3	+51	-0.1369	+0.0167

5. Why ODD+game Was Not Always Better

The game scenario adds useful institutional context, but it also increases the amount of salient mechanism text available to the model. Some models treat this background as additional ASs rather than as explanatory context. The AS-level audits show three main failure modes: duplicate extraction of a correct AS, over-splitting background mechanisms into standalone ASs, and missing a specific correct AS because the model merges it into a broader mechanism.

Table 5. Main reasons ODD+game was not always better

Case	Model	Where ODD+game underperformed	Audit-based explanation
Water-use	DeepSeek-R1	Precision slightly lower	ODD+game improved recall (+0.0500) and TP (+3), but FP increased by 1. Extra outputs included lake-threshold and fishery-sustainability variants.
Water-use	Llama-3.3-70B	Both precision and recall lower	ODD+game had -1 TP, +1 FN, and +9 FP. It over-generated ecological-threshold, irrigation-investment, field-expansion, and fishing-quota themes, and missed the fish extraction AS six times.
Electricity	Llama-3.3-70B	Recall lower	ODD+game had -10 TP and +10 FN, mostly because it missed the sequential social-learning capacitor-adoption AS 11 times. The added context likely made the model merge learning into capacitor adoption instead of extracting it separately.
Electricity	Qwen2.5-7B	Precision lower	ODD+game reached perfect recall but FP increased by 51. Most FP were duplicates, with some wrong AS7-AS10 style outputs such as bounded rationality, grid reliability, and social norms.

6. Interpretation

- ODD+game is most useful when the added scenario clarifies the correct strategic structure without inviting new standalone AS labels.
- DeepSeek-R1 benefited most clearly in the electricity case: TP increased from 164 to 180, FN dropped from 16 to 0, and FP dropped from 12 to 4.
- For water-use, the hard AS is AS2, the fish extraction common-pool resource game. All models missed it more often than AS1, and many FP-wrong titles drifted into ecological-threshold or fish-reproduction themes.
- For electricity, Qwen2.5-7B showed the classic high-recall, low-precision pattern: it found every correct AS under ODD+game, but it generated many duplicate or extra ASs.

- The most trustworthy quality judgment should therefore use both aggregate metrics and AS-level audit files, not just whether ODD+game has higher recall.

7. Source Artifacts

Artifact type	Path
Water comparison report	ODD_water/ Water_evaluation_comparison_ODD+game_stuff_vs_ODD-only.txt
Water AS-level audit, ODD+game	ODD_water/Batch_30Runs/ Water_evaluation_as_level_ODD+game_stuff.csv
Water AS-level audit, ODD-only	ODD_water/Batch_30Runs_ODDOnly/ Water_evaluation_as_level_ODD-only.csv
Electricity comparison report	ODD_electricity/ Electricity_evaluation_comparison_ODD+game_stuff_vs_ODD-only.txt
Electricity AS-level audit, ODD+game	ODD_electricity/Batch_30Runs/ Electricity_evaluation_as_level_ODD+game_stuff.csv
Electricity AS-level audit, ODD-only	ODD_electricity/Batch_30Runs_ODDOnly/ Electricity_evaluation_as_level_ODD-only.csv